RooftopSolar Suite v2.5

Complete AutoCAD Engineering User Guide & Command Reference Manual

Platform: AutoCAD 2018–2027 • Architecture: AutoLISP + DCL • Edition: PVDesigner 2027 Enhanced

1. Distribution Package & Files to Share

When deploying or sharing the RooftopSolar application with engineering colleagues, CAD drafters, or EPC teams, always ensure that the files remain inside the same directory. The AutoLISP suite uses automated relative path detection so the software can be extracted anywhere on any workstation without modifying system search paths.

Filename	Type	Functional Role & Details
RooftopSolar.lsp	AutoLISP Engine	Core computational engine containing 56 engineering commands, polyline boundary solvers, obstacle shadow casting, DC stringing algorithms, cable sizing, SLD generation, and yield math.
RooftopSolar.dcl	Dialog Definition	CAD interface definitions for the Master Functions Control Center (PVMENU) and the 6-Tab Interactive SolarTools Designer (ROOFTOPSOLAR).
PVCompLib.lsp	Component Library	Persistent database storing Tier-1 modules and multi-MPPT inverters. All imported .PAN and .OND files automatically save here for permanent cross-project use.
USER_GUIDE.md	Documentation	Complete user manual, command cheatsheet, and quick start reference guide.
***.PAN / *.OND**	Sample Specs	Real-world PVsyst equipment files (JinkoSolar 550W, LONGi 550W, Trina 550W, Sungrow 50kW, Solis 50kW) for testing imports.
RooftopBoundary.dwg	Sample Drawing	Ready-to-use practice rooftop model with parapet walls, skylights, and obstructions for verification.

> **[MANDATORY REQUIREMENT]** The three core files (RooftopSolar.lsp, RooftopSolar.dcl, and PVCompLib.lsp) must stay in the same folder. If separated, AutoCAD will not be able to load the dialog interfaces or persistent equipment libraries.

2. Installation & Quick Setup

Method A: Quick Loading per Session

1. Open AutoCAD (versions 2018 through 2027 are fully supported).

2. Type APPLOAD in the command bar and press Enter.

3. In the file dialog, navigate to the folder where you extracted the package, select RooftopSolar.lsp, and click Load.

4. AutoCAD will report: 'Solar PV Plant Layout Generator - Rooftop Projects Loaded!'

Method B: Permanent Auto-Load via Startup Suite (Recommended)

1. Type APPLOAD in the command bar.

2. Under the 'Startup Suite' briefcase section in the bottom right, click Contents...

3. Click Add..., select RooftopSolar.lsp, and click Open.

4. Click Close. RooftopSolar will now automatically load whenever any AutoCAD drawing is opened.

> **[CRITICAL DRAWING UNITS RULE]** The suite uses real-world SI engineering coordinates where 1 AutoCAD Drawing Unit = 1 Meter. Type UNITS in AutoCAD and ensure Insertion Scale is set to 'Meters' and Length Type is 'Decimal'. A 30m x 20m building should be drawn as 30 x 20 units in CAD.

3. Primary User Interfaces

RooftopSolar offers two convenient entry points depending on whether you prefer a step-by-step wizard or an instant one-click command dispatcher:

Interface	Command / Alias	Operational Purpose
Master Functions Hub	PVMENU / PVDCL	One-window control center displaying all 56 engineering functions neatly grouped into categories with descriptions. Click any button to run.
Full Designer Wizard	SOLARTOOLS / ROOFTOPSOLAR	Comprehensive 6-Tab guided wizard (Site & Roof, Equipment, Array Layout, Walkways & Safety, Electrical & 3D, Financial ROI & BOM).
Direct Command Line	e.g. PVSTRING, PVSLD	All 56 commands can be executed directly from the AutoCAD command line or mapped to custom tool palettes.

4. End-to-End 10-Stage Engineering Workflow

Stage 1: Define Roof Boundaries & 3D Building Outlines (PVPICKBOUNDARY / PVBLDG3D)

Draw the building perimeter as a closed polyline (PLINE). Run PVPICKBOUNDARY to select single or window-select multiple closed polylines. For multi-building campuses, run PVMULTIROOF to inspect individual roof handles and areas (m^2), and PVSELECTROOF to toggle active roofs. The suite immediately prompts for Rooftop Type [Flat-RCC / Metal-Sheet]. For Metal Sheet, configure 1, 2, or 4 slope directions, define slope facing orientation (North/South/East/West for 1-dir; North & South vs East & West for 2-dir), and input building heights at the centre (ridge) and ends (eaves). The suite computes the true slope angle and creates 3D vertical walls (PV_3D_BUILDING), eaves and ridge outlines (PV_3D_ROOF_OUTLINE), and roof surfaces (PV_3D_ROOF_FACE). Reconfigure anytime with PVBLDG3D.

Stage 2: Complex Roof Splitting (PVHIPSPLIT / PVDOMESPLIT)

For residential or pitched commercial sites: Run PVHIPSPLIT to automatically bisect hip-and-valley roofs along ridgelines into distinct planar faces. For curved cylindrical barrel vaults or dome roofs, run PVDOMESPLIT to discretize curvature into flat facet strips, or PVONEDOME to normalize slopes into an equivalent coordinate plane.

Stage 3: Rooftop Obstacles & Solar Shadows (PVPICKOBSTACLES)

Select obstructions (HVAC units, skylights, water tanks, elevator headrooms) drawn as closed polylines. Specify obstacle height (e.g. 1.5m). The suite automatically projects winter-solstice solar shadow envelopes on layer 'PV_SHADOW' based on site latitude. Modules automatically avoid shaded areas.

Stage 4: Equipment Database & Persistent Library (PVCOMPLIB)

Run PVLOADPAN to import any manufacturer PVsyst .PAN file. Electrical ratings (Vmp, Imp, Voc, Isc, temp coefficients, bifaciality) are parsed and auto-saved to PVCompLib.lsp. Run PVLOADOND to import string inverters (.OND) with MPPT voltage tracking windows and DC capacity ratings. Run PVCOMPLIB to view, reload, or manage your equipment database.

Stage 5: High-Density Array Layout (PVLAYOUT / PVLAYOUTEW)

Choose between 'Max Polyline Fill' or 'Target Plant Capacity (kWp)' in Tab 1. Run PVLAYOUT to populate tables respecting setbacks, inter-row pitch, and obstacles. For dual-tilt East-West (A-frame / butterfly) configurations, run PVLAYOUTEW to place dual-tilt East (90°) and West (270°) subarrays with dedicated layer separation.

Stage 6: Maintenance Walkways & Safety Systems (PVWALKWAY / PVFALLSAFETY)

Run PVWALKWAY to draw maintenance corridors (default 0.80m width) on layer 'PV_WALKWAY'. Run PVFALLSAFETY to insert EN 795 / OSHA compliant wire-rope lifelines with spring tensioners and corner anchors on 'PV_FALL_SAFETY'. Run PVHANDRAIL to insert perimeter dual-rail safety handrails with stanchion posts on 'PV_HANDRAIL'. All safety paths act as exclusion zones during layout.

Stage 7: Boustrophedon DC Stringing & Numbering (PVSTRING / PVSTRINGMANUAL)

Run PVSTRING and input target modules per string (e.g. 20). The boustrophedon snake sorting algorithm connects modules row-by-row in alternating directions to minimize DC cable loop area. Polylines are drawn in rotating vivid colors on 'PV_STRINGS', stamping [+] Red and [-] Blue terminal poles. String tags are created on 'PV_STRING_TAGS'. Use PVSTRINGMANUAL for custom click-to-wire layouts, PVSTRINGEW for isolated East vs West series circuits, PVPOLARITY to invert poles, and PVPANELNUM to serialize modules ('STR-01-01').

Stage 8: Inverter Plinth, MPPT Linking & Cable Trays (PVINVERTER / PVCABLETRAY)

Run PVINVERTER to place a concrete plinth, inverter enclosure, DC/AC switches, and engineering rating tag on 'PV_INVERTER'. Run PVMPPTDETAILS to view voltage windows, and PVMPPTLINK to wire string terminals to inverter MPPT inputs. Run PVCABLETRAY to route a perforated GI cable tray corridor with rungs. Run PVDCROUTETRACKED to route home-runs along the tray, and PVCABLECALC to size conductors and verify %Vdrop < 1.5%.

Stage 9: 3D Visualization & Shade Pruning (PV3D / PVPRUNESHADED)

Run PV3D to convert 2D modules into tilted 3DFACE planes, extrude MMS support legs and purlins, and extrude obstacles into 3D solid blocks. Modules automatically sit on top of the 3D building at their true roof elevations. Automatically switches to South-West Isometric with Conceptual Shading. Run PVPRUNESHADED to audit modules against winter solstice shadow profiles and auto-delete shaded modules. Run PVCLEAR3D to return to 2D.

Stage 10: Deliverables & Export (PVSLD / PVENERGYTABLE / PVPVSYSTEXPORT)

Run PVSLD to generate an IEC 62446-1 Single Line Diagram schematic. Run PVENERGYTABLE for an AutoCAD native 25-Year yield table. Run PVDRAWBOM for a quantified Bill of Materials, PVDCSCHEDULE for a DC cable schedule, PVSIMREPORT for a simulation text report, and PVPVSYSTEXPORT for a native PVsyst 7/8 3D scene (.PVC) export.

5. Master Command Cheatsheet (Alphabetical)

Command / Aliases	Category	Functional Description
PV3D / PV3DVIEW	3D Modeling	Generate 3D tilted module planes atop building, MMS legs, 3D obstacle solids, and South-West Isometric conceptual view.
PVACEQUIPMENT	Balance of System	Place AC Distribution Boards (ACDB), Main Switchboards (MSB), and transformer symbols with tags.
PVBLDG3D / PVBUILDING3D	3D Building	Interactive 3D building outlines modeler: configure roof slope (1/2/4 dir), facing orientation & heights.
PVCABLECALC	Calculations	BS 7671 / IEC 60364-7-712 DC & AC solar cable cross-section (4/6/10 mm²) and %Vdrop sizing calculator.

Command / Aliases	Category	Functional Description
PVCABLETRAY	Electrical	Route perforated GI cable tray corridor with automatic cross-rungs and length accumulation.
PVCLEAR3D	3D Modeling	Delete 3D tilted surfaces, MMS legs, and obstacle blocks, returning viewport to clean 2D plan view.
PVCLEARCABLETRAY	Electrical	Erase all cable tray corridors and associated guide elements.
PVCLEARLAYOUT	Layout	Erase placed solar modules, MMS table structures, and array annotations.
PVCLEAROBSTACLES	Obstacles	Remove registered obstacle boundaries and winter solstice shadow envelopes.
PVCLEARSTRINGS	Electrical	Delete all DC string circuit polylines, [+] / [-] terminal markers, and string tags.
PVCOMPLIB / PVLIBRARY	Equipment	Interactive component library manager: inspect saved modules & inverters, reload, or force save.
PVDCSCHEDULE	Deliverables	Generate AutoCAD native DC Cable Schedule Table documenting strings, homerun lengths, and %Vdrop.
PVDCROUTETRACKED	Electrical	Route DC homerun conduits snapped along designated cable tray corridors.
PVDIMPLAN	Drafting	Automated dimensioning of table pitch, row spacing, boundary setbacks, and walkway corridors.
PVDOMESPLIT	Roof Modeling	Discretize curved barrel vault or dome roof polylines into flat planar facet strips.
PVENERGYTABLE	Deliverables	Insert publication-ready AutoCAD native 25-Year Yield & Monthly Energy Generation Table.
PVEWVIS	Layout	Interactively toggle visibility of East-facing vs West-facing subarrays in dual-tilt layouts.
PVEXPORTCSV	Deliverables	Export monthly yield simulation and 25-year degradation figures to standard CSV.
PVFALLSAFETY	Safety Systems	Draw EN 795 / OSHA wire-rope lifelines with spring-loaded tensioners and corner anchors.

Command / Aliases	Category	Functional Description
PVHANDRAIL	Safety Systems	Draw perimeter dual-rail safety guardrails with stanchion posts and kickplates.
PVHIPSPLIT	Roof Modeling	Bisect hip and valley roof polylines along ridgelines into distinct planar faces.
PVINVERTER	Electrical	Place inverter concrete plinth, enclosure, DC/AC switches, and engineering rating plate.
PVLAYOUT	Layout	Populate solar PV module tables inside boundary (Max Polyline Fill or Target kWp mode).
PVLAYOUTEW	Layout	Generate dual-tilt East-West (A-frame / butterfly) array layout with series-isolated layers.
PVLOADOND / PVOND	Equipment	Browse and parse PVsyst inverter file (.OND), saving parameters permanently to PVCompLib.lsp.
PVLOADPAN / PVPAN	Equipment	Browse and parse PVsyst module file (.PAN), saving electrical specs permanently to PVCompLib.lsp.
PVMENU / PVDCL	Navigation	Open Master Functions Control Center dialog with direct execution buttons for all tools.
PVMPPTDETAILS	Electrical	Display inverter MPPT voltage windows, tracker allocations, and maximum string limits.
PVMPPTLINK	Electrical	Route electrical tie lines from string terminal poles directly to designated inverter MPPT inputs.
PVMULTIROOF / PVROOFS	Roof Management	Manage multi-building and multi-zone roof campuses with interactive roof list.
PVONEDOME	Roof Modeling	Convert faceted cylindrical barrel vault sections into a single normalized coordinate plane.
PVPANELNUM	Electrical	Sequential module numbering: stamp unique IDs on layer PV_PANEL_NUMS ('STR-01-01'...).
PVPICKBOUNDARY	Layout	Pick single or window-select multiple closed polyline roof outlines.

Command / Aliases	Category	Functional Description
PVPICKOBSTACLES	Obstacles	Select rooftop obstructions and project winter-solstice solar shadow envelopes.
PVPOLARITY	Electrical	Reverse circuit polarity on selected string, swapping [+] and [-] poles to shorten homerun runs.
PVPRUNESHADED	3D / Shading	Highlight over-shaded modules in bright red and auto-delete to protect array PR.
PVPVSYSTEXPORT	Export	Export complete 3D array coordinate layout into native PVsyst 7/8 scene (.PVC) format.
PVROOFLIST	Roof Management	Display inventory list of all detected roofs with handles, vertex counts, and areas (m²).
PVROOFNUM	Roof Management	Stamp circular badges and net usable roof area annotations at roof centroids.
PVSELECTROOF	Roof Management	Select a specific active target roof (1..N) or set target to ALL ROOFS (0).
PVSIMREPORT	Deliverables	Generate comprehensive IEC 61724 technical and financial simulation report in Notepad.
PVSLD / PVDRAWSLD	Electrical	Auto-generate publication-ready IEC 62446-1 Single Line Diagram (SLD) schematic drawing.
PVSTRING	Electrical	Automatic boustrophedon snake DC stringing with colored circuits and [+] / [-] terminals.
PVSTRINGEW	Electrical	East-West stringing: series strings East and West subarrays in strictly isolated circuits.
PVSTRINGMANUAL	Electrical	Interactive click-to-wire manual stringing for irregular roofs or custom array shapes.
PVSTRINGVIS	Electrical	Convex-hull string cluster visualizer displaying colored boundary groups for QA audit.
PVVALIDATE	QA / Audit	Comprehensive pre-flight engineering validation of setbacks, MPPT voltages, and shading risks.

Command / Aliases	Category	Functional Description
PVWALKWAY	Walkways	Draw maintenance walkway corridors (custom width, e.g. 0.80m) on layer PV_WALKWAY.
SOLARTOOLS / ROOFTOPSOLAR	Navigation	Launch full 6-Tab Interactive Solar PV Designer Wizard.

6. Persistent Component Library (PVCompLib) Architecture

PVCompLib solves the persistent problem of losing imported equipment specifications across different project files. Whenever a user imports a .PAN or .OND file, the suite parses all properties and writes them directly into PVCompLib.lsp located in the application directory:

• Automatic Save: Runs silently in the background whenever a new module or inverter is loaded.

• Automatic Load: Runs automatically whenever RooftopSolar.lsp is loaded into any new or existing drawing.

• Deduplication: Replaces existing entries if the same model name is re-imported, updating specs seamlessly.

• Portable AutoLISP Format: Uses native LISP data structures (assoc-lists), requiring no external database engines.

• Command Management: Run PVCOMPLIB to inspect the database, reload from disk, or open the file in Notepad.

7. Troubleshooting & FAQ

Dialog Does Not Appear ('Could not find RooftopSolar.dcl')

Ensure RooftopSolar.dcl is located in the exact same directory as RooftopSolar.lsp. The suite automatically checks its own load directory first. If files were separated, place them together or add the directory to AutoCAD's Support File Search Path (OPTIONS > Files > Support File Search Path).

Imported Equipment Not Saving on AutoCAD Exit

Ensure the folder where RooftopSolar is stored has Windows write permissions. Avoid placing the application folder inside 'C:\Program Files\' where Windows restricts file writing without administrator elevation. Place it in an accessible location such as 'D:\SolarTools\' or 'C:\Users\<Name>\Documents\AutoCAD\SolarTools\',

Modules Placed at Unexpected Angles or Outside Roof Boundary

1. Verify that your roof boundary is a strictly Closed 2D Polyline (in the AutoCAD Properties palette, ensure 'Closed = Yes').
2. Verify drawing units with the UNITS command. Ensure units are set to Meters (not millimeters). A 30m building must measure 30 units, not 30,000 units.

DC Strings Not Connecting All Modules in a Row

Check the modules-per-string setting. If a row contains 25 modules and string size is 20, the algorithm completes string 1 (20 modules) and begins string 2 with the remaining 5 modules, continuing into the adjacent row. This is standard boustrophedon stringing topology.